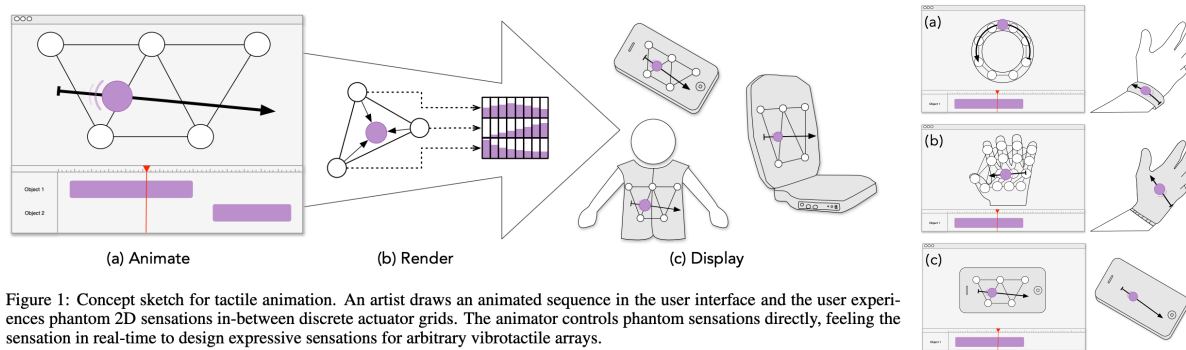


Tactile Animation

We introduce the tactile animation object, a directly manipulated phantom tactile sensation. This abstraction has two key benefits: 1) efficient, creative, iterative control of spatiotemporal sensations, and 2) the potential to support a variety of tactile grids, including sparse displays. We present Mango, an editing tool for animators, including its rendering pipeline and perceptually-optimized interpolation algorithm for sparse vibrotactile grids. In our evaluation, professional animators found it easy to create a variety of vibrotactile patterns, with both experts and novices preferring the tactile animation object over controlling actuators individually.



- Schneider, O. S., Israr, A., & MacLean, K. E. (2015, November). Tactile animation by direct manipulation of grid displays. In *Proceedings of the 28th annual ACM symposium on user interface software & technology* (pp. 21-30). [\[Link\]](#)

▶ Tactile Animation by Direct Manipulation of Grid Displays